

The Fundamentals of Programming

Booleans & Conditional Statements



Objectives

What you will learn:

1. How to work with Boolean values.
2. How to use conditional statements to make decisions.
3. The ways that you can put together different conditional statements to make complex decisions.

Boolean values

Truthiness

In Python there is another type of variable called Boolean. All Boolean variables will contain either **True** or **False**.

Look at the expressions to the right and decide if they are **True** or **False**.

`5 * 10 < 300`

`10 % 3 >= 1`

`9 ** 2 <= 50`

`pow(2, 3) <= 8`

`min(1, 2, 3, 4) * 10 > 9.9`

Comparison operators

Other equals?

Most operators are similar to math and pretty easy to figure out:

< <= >= >

But these two are different:

== equal

!= not equal

56 == 56.0

42 != 42

max(1, 3, 5) == pow(5, 1)

7 * 8 % 3 != 7 % 8 * 3

100 - 2 == 48 * 2

The Equality Operator

It's important to remember in Python the different between the assignment operator = and the equality operator ==.

Look at this example:

```
age = 10  
print(age)  
age == 10  
print(age)  
print(age == 10)
```

Remember that the first line is:
“age is assigned the value 10”

What will the output of the above program be?

Conditionals

If statements

If statements can have one or more sections, and can be used to make decisions. With if statements you:

- **must use logical operators** (<, >, ==, etc...)
- **only choose one “branch”**
- **first True branch is chosen**

While the first **if** is necessary, both the **elif** and **else** are optional.

```
age = int(input())
```

```
if age > 50:
```

```
    print("You're too old")
```

```
elif age > 12:
```

```
    print("You're old enough!")
```

```
else:
```

```
    print("You're too young!")
```

Practice

Basic conditionals

For both of these programs complete the conditional statements.

100 NT is required

```
money = int(input("You have: "))
```

```
-----:
print("Not enough")
```

```
-----:
print("Enough")
```

50NT for kids 100NT for Adults

```
money = int(input("You have: "))
```

```
-----:
print("Adult ticket?")
```

```
-----:
print("Kid ticket?")
```

```
-----:
print("Not enough")
```

Conditionals combined

and, or, not

Booleans have special operators called **logical operators** which are: **and**, **or**, and **not**.

In looking at the programs to the right, what do you think the output will be?

```
print(5 < 6 and 2 > 1)
print(1 < 0 and 2 > 3)
print(3 == 3 or 1 != 3)
print(5 < 1 or 2 > 9)
print(not True)
print(not False)
print(not(5 < 6))
```


Logicals Operators

With logical operators you can combine as many conditional statements as you want.

Look at this example:

```
age = int(input())  
height = int(input())
```

if age > 10 and height > 150:

```
    print("You're tall for your age!")
```

elif age >= 16 and height < 150:

```
    print("You'll grow into it.")
```

elif age == 15 or height != 200:

```
    print("Weird check, but meh.")
```

You could continue and check for any number of circumstances.

Practice

Conditional statements

Determine the final out of these programs.

```
a = 10
b = 4
if (a < 4) or (b**2) > a:
    a = a * 2
elif b < a:
    a = a + 1

print(a)
```

```
a = 10
b = 4
if (b == 4) and (b**2 != 16):
    a = a * 2
elif b < a and a // 3 != 3:
    a = a + 1

print(a)
```

Practice

With functions

Write a function that takes in two points and returns the distance between them, but only if the distance is more than 10, otherwise it returns -1.

```
def dist(x1, y1, x2, y2):  
    #Do work here
```

```
print(dist(0, 0, 12, 0))  
print(dist(0, 0, 9, 0))  
print(dist(-12, 3, 14, 50))
```

Recap

We learned

1. That Booleans types can hold **True** or **False**.
2. How to use logical operators `>`, `<`, `>=`, `<=`, `!=`, and `==`
3. To create conditional statements using **if**, **else if**, and **else**.
4. How to combine conditional statements use logical operators **and**, **or**, and **not**.